

1a

given analog signal:

$$x(t) = 2 \sin 300\pi t + 5 \sin 500\pi t$$

$$= 2 \sin 2\pi 150t + 5 \sin 2\pi 250t$$

$$A_1 = 2, \quad f_1 = 150 \text{ Hz}, \quad A_2 = 5, \quad f_2 = 250 \text{ Hz}$$

$$f_{\max} = 250 \text{ Hz}$$

$$\text{i) } \therefore \text{ Nyquist rate } \geq 2 \times 250$$

$$\geq \underline{500 \text{ samples/sec}}$$

$\therefore$ minimum sampling rate required to avoid aliasing is 500 samples/sec.

$$\text{ii) } f_s = 200 \text{ Hz}$$

$$T = \frac{1}{200}$$

discrete time signal obtained after sampling =

$$x(n) = 2 \sin 2\pi 150 \frac{n}{200} + 5 \sin 2\pi 250 \frac{n}{200}$$

$$\therefore x(n) = 2 \sin 2\pi \frac{3n}{4} + 5 \sin 2\pi \frac{5n}{4} \quad \#$$

iii) given,

$$0 < F < f_s/2$$

$$0 < F < 100$$

now, using formula

$$F = f_s - Kf_s$$

$$f_1 = f_s - Kf_s \\ = 150 - 1 \times 200 = -50 \quad (K=1)$$

$$f_2 = 250 - 1 \times 200 = 50$$

since $0 < 50 < 100$

$\therefore$ The frequency identical for part (ii)
 $= 50 \text{ Hz.}$

1b

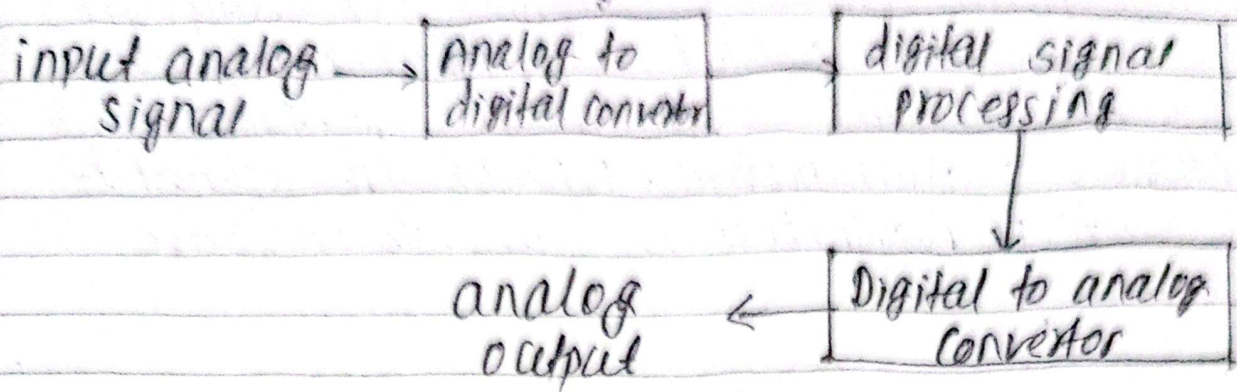


fig: block diagram of digital signal processing

Digital signal processing is the mechanism to find required data from signal or manipulation of signal. It has 5 steps as:

a) Input analog signal

- In this step we give a analog signal as input in a system. It is initial state.

b) Analog to digital converter

- After obtaining of analog signal as input a A/D converter converts the signal into digital signal. It is sampling where continuous signal changed to digital signal by a sampler.

c) Digital Signal Processing

- It is main part of signal processing. In this state useful information or data of digital signal is extracted. We analyse, manipulate, update the signal at it state. $\therefore$ Amplifier is used to increase frequency etc.

d) Digital to Analog Converter

After gaining useful information of digital signal now D/A converter changes it to analog signal. Discrete signal changes to continuous form.

e) Analog output signal

After whole processing finished we get analog signal as output. Which is displayed by various devices.

2a

- Discrete-time signals are the signals which are defined for fixed point of time.
- It is a digital signal.
- Discrete signal can be periodic, aperiodic or even odd.

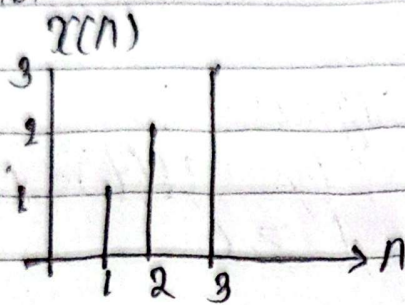


fig: discrete signal

Elementary signals are the signals which are used to build block of signal to analyse complex signals. Elementary signals are as follows:

a) Unit Impulse signal

- It is denoted by $\delta(t)$ or $\delta(n)$ is defined at $t=0$ only.

$$\delta(t) = \begin{cases} 1 & \text{for } t=0 \\ 0 & \text{else} \end{cases} \quad // \text{continuous signal}$$

$$\delta(n) = \begin{cases} 1 & \text{for } n=0 \\ 0 & \text{else} \end{cases} \quad // \text{discrete signal}$$

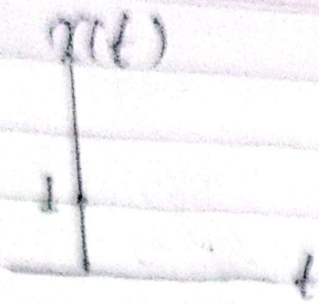


fig: $\delta(t)$

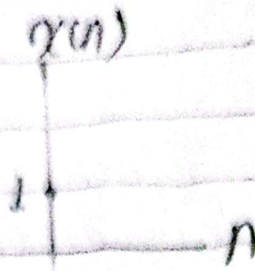


fig: $\delta(n)$

b) Unit - step signal

- It is denoted by $u(t)$ or $u(n)$ and define at $t \geq 0$.

$$u(t) = \begin{cases} 1 & \text{if } t \geq 0 \\ 0 & \text{else} \end{cases}$$

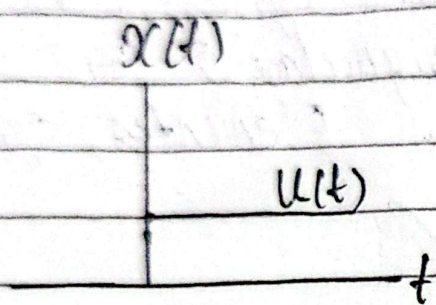


fig: $u(t)$

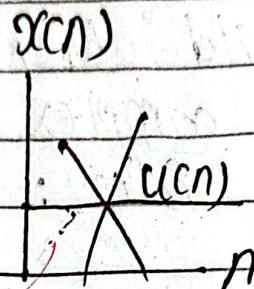
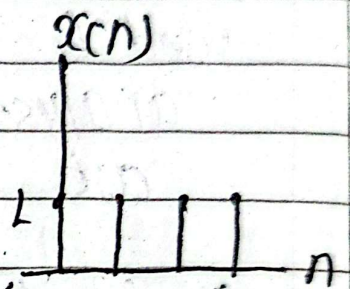


fig: $u(n)$



c) Ramp Signal

- It is denoted by $r(t)$ or $r(n)$ and increases linearly with time.

$$r(t) = \begin{cases} t & \text{if } t \geq 0 \\ 0 & \text{else} \end{cases}$$

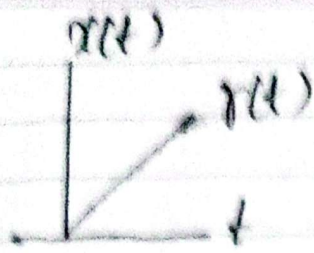


fig: $x(t)$

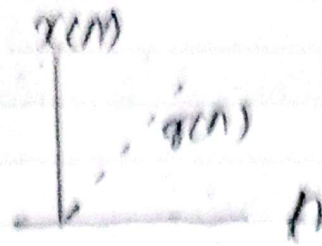


fig: $x(n)$

d) Signum Signal

- It is denoted by $\text{sign}(t)$ or $\text{sign}(n)$ and defined for all value except $t = 0$

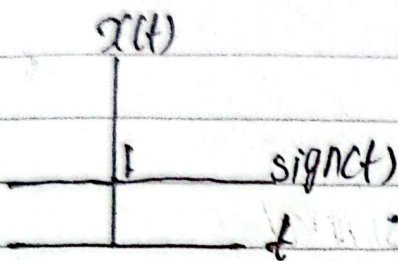


fig: $\text{sign}(t)$

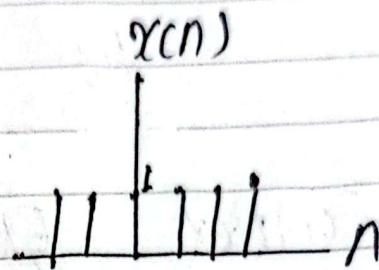


fig: $\text{sign}(n)$

$$\text{sign}(t) = \begin{cases} 1 & \text{for } t \neq 0 \\ 0 & \text{for } t = 0 \end{cases}$$

e) sinc signal

- It is denoted by $\text{sinc}(t)$ or $\text{sinc}(n)$ and defined as $\frac{\sin t}{t}$ for $t \neq 0$.

- It is value 1 as peak at $t = 0$.

- It L Hospital's law

$$\lim_{t \rightarrow 0} \frac{\sin t}{t} = 1$$

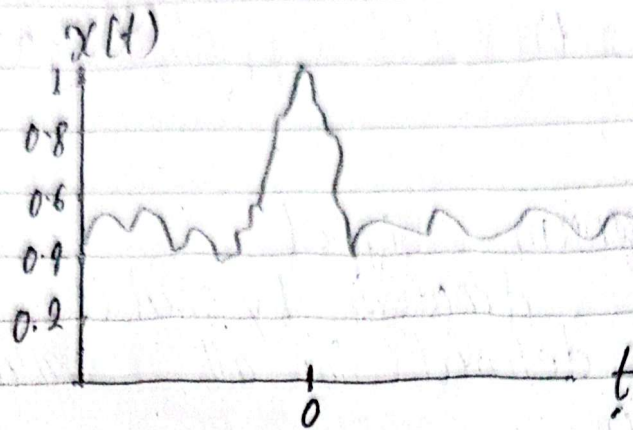


fig: $\text{sinc}(t)$

f) Complex Exponential signal

- It is defined as

$$Ae^{j\omega t}$$

where A and j is complex number.

∴ It is defined for all point $t = 0$, -ve and +ve value.

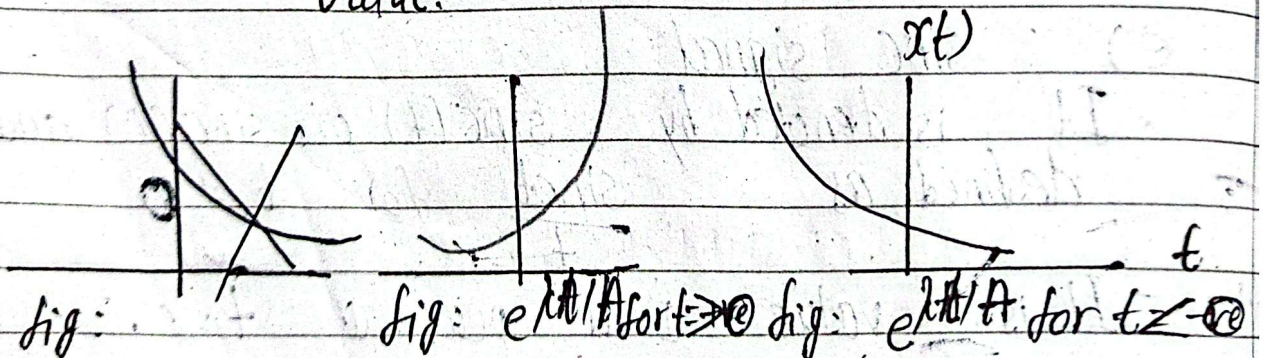


fig:

fig:

fig:

Qb

$$x_1(n) = \{1, 2, 5, 6\}, \quad x_2(n) = \{2, 1, 0, 2\}$$

$$x_1(0) = 1$$

$$x_1(1) = 2$$

$$x_1(2) = 5$$

$$x_1(3) = 6$$

$$x_2(0) = 2$$

$$x_2(1) = 1$$

$$x_2(2) = 0$$

$$x_2(3) = 2$$

limit = 0 to 6

$$\begin{aligned} \text{convolution sum} &= x_1(n) * x_2(n) = \\ y(n) &= \sum_{k=0}^6 x_1(k) x_2(n-k) \end{aligned}$$

$$n=0,$$

$$y(0) = \sum_{k=0}^6 x_1(k) x_2(0-k)$$

$$= x_1(0) x_2(0) + x_1(1) x_2(-1) + x_1(2) x_2(-2) + x_1(3) x_2(-3)$$

$$0 = 1 \times 2 = 2$$

$$y(1) = \sum_{k=0}^6 x_1(k) x_2(1-k)$$

$$= x_1(0) x_2(1) + x_1(1) x_2(0)$$

$$= 1 \times 1 + 2 \times 2 = 5$$

$$\begin{aligned}
 Y(2) &= \sum_{k=0}^2 x_1(k) x_2(2-k) \\
 &= x_1(0)x_2(2) + x_1(1)x_2(1) + x_1(2)x_2(0) \\
 &= 1 \times 0 + 2 \times 1 + 5 \times 2 \\
 &= 12
 \end{aligned}$$

$$\begin{aligned}
 Y(3) &= x_1(0)x_2(3) + x_1(1)x_2(2) + \\
 &\quad x_1(2)x_2(1) + x_1(3)x_2(0) \\
 &= 1 \times 2 + 2 \times 0 + 5 \times 1 + 6 \times 2 \\
 &= 19
 \end{aligned}$$

$$\begin{aligned}
 Y(4) &= x_1(0)x_2(4) + x_1(1)x_2(3) + \\
 &\quad x_1(2)x_2(2) + x_1(3)x_2(1) \\
 &= 1 \times 0 + 2 \times 2 + 5 \times 0 + 6 \times 1 \\
 &= 10
 \end{aligned}$$

$$\begin{aligned}
 Y(5) &= 1 \times 0 + 2 \times 0 + 5 \times 2 + 6 \times 0 \\
 &= 10
 \end{aligned}$$

$$\begin{aligned}
 Y(6) &= 1 \times 0 + 2 \times 0 + 5 \times 0 + 6 \times 2 \\
 &= 12
 \end{aligned}$$

$$\therefore Y(n) = \{2, 5, 12, 19, 10, 10, 12\}$$

3a.

- System is a device in which we give a input and it gives corresponding output.

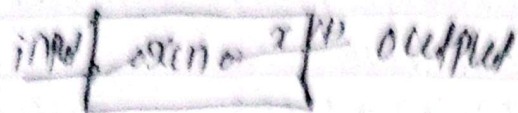


fig: system

- Eg: Amplifiers, transistor, MUX, DEMUX etc.
- a single input can produce different output varies with time and environment in a system.

Property of Systems:

a) Linearity

A system is called linear if it satisfies superposition theorem else not linear.

✓ $y(n)$ is function of x then.

$$a_1 x_1(n) + a_2 x_2(n) = a_1 y_1(n) + a_2 y_2(n)$$

if the system doesn't satisfies it, then that is non-linear.

b) Causality

A system is called causal if its output depends on present and past value. System is non-causal if it depends on present, past as well as future value.

$$x(n) = u(n-1) + u(n) \quad \text{— causal.}$$

$$x(n) = u(n+1) + u(n-1) \quad \text{— non causal}$$

c) Time Invariant

A signal is time invariant if input output system invaries with time else time variant.

$$x(n) = 3n^2 - 6 \quad \text{is not time invariant.}$$

— small shift in time not shift output.

d) Stability

A system is stable if bounded input produces bounded output.

$$x(n) = 3n^2 - 6 \quad \text{is stable signal.}$$

— unstable has unbounded value like ∞ .

e) Dynamic / static

A system is static if it depends only on current value and no memory is needed and dynamic if it depends on current value as well as past value and has memory.

$$x(t) = 3x^2 + 6 \quad \text{= static}$$

$$x(t) = 3(x-3)^2 + 6 \quad \text{= dynamic}$$

Qb

$$\begin{aligned}x(n) &= u(n) \\ h(n) &= a^n u(n)\end{aligned}$$

$$\text{response } y(n) = x(n) * h(n) = \sum_{k=0}^{\infty} h(k) x(n-k)$$

$$\text{as, } u(n) = \begin{cases} 1 & \text{for } n \geq 0 \\ 0 & \text{for else} \end{cases}$$

$$\begin{aligned}y(n) &= h(0)x(n) + h(1)x(n-1) + \\ & h(2)x(n-2) + h(3)x(n-3) + \dots \\ &= \cancel{a^0 u(n) x(n)} \\ &= a^0 x(n) + a^1 x(n-1) + a^2 x(n-2) + \dots\end{aligned}$$

$$\therefore y(n) = 1 + a + a^2 + a^3 + \dots$$

$$y(n) = \sum_{k=0}^{\infty} a^k$$

$$= \sum_{N_1}^{N_2} a^N = \frac{a^{N+1} - a^{N_1}}{N+1 - N_1} \quad \#$$

$$\text{where } N_1 = 0, N_2 = \infty$$

5

4a

$$x(n) = a^n u(n), \quad a > 0$$

$$x(0) = a^0 u(0) = 1$$

$$x(1) = a$$

$$x(2) = a^2$$

$$x(3) = a^3$$

$$Z[x(n)] = X(z) = \sum_{n=0}^{\infty} x(n) z^{-n}$$

$$= x(0)z^{-0} + x(1)z^{-1} + x(2)z^{-2} + x(3)z^{-3} + \dots$$

$$= 1 + az^{-1} + a^2z^{-2} + a^3z^{-3} + \dots$$

$$= 1 + \frac{a}{z} + \frac{a^2}{z^2} + \frac{a^3}{z^3} + \dots$$

$$= \sum_{n=0}^{\infty} \left(\frac{a}{z} \right)^n$$

$$= \frac{1}{1 - a/z} = \frac{z}{z - a}$$

$$\therefore Z[x(n)] = \frac{z}{z - a} \quad \#$$

$$\text{ROC} = |z| > a$$



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4b

$$\begin{aligned}x(n) &= \cos(\omega_0 n) u(n) \\&= \cos(\omega_0 n) \quad n \geq 0\end{aligned}$$

$$\therefore Z[x(n)] = X(z) = \sum_{n=0}^{\infty} x(n) z^{-n}$$

~~4b~~

$$\cos(\omega_0 n) = \frac{e^{j\omega_0 n} + e^{-j\omega_0 n}}{2}$$

$$= \frac{1}{2} (e^{j\omega_0 n} + e^{-j\omega_0 n})$$

$$= \frac{1}{2} [(e^{j\omega_0})^n + (e^{-j\omega_0})^n]$$

$$Z[\cos \omega_0 n] = \frac{1}{2} Z[(e^{j\omega_0})^n + (e^{-j\omega_0})^n]$$

$$= \frac{1}{2} \left(\frac{z}{z - e^{j\omega_0}} + \frac{z}{z - e^{-j\omega_0}} \right)$$

$$= \frac{1}{2} \times \frac{z(z - e^{-j\omega_0}) + z(z - e^{j\omega_0})}{(z - e^{j\omega_0})(z - e^{-j\omega_0})}$$

$$= \frac{1}{2} \times \frac{z^2 - ze^{-j\omega_0} + z^2 - ze^{j\omega_0}}{(z - e^{j\omega_0})(z - e^{-j\omega_0})}$$

$$= \frac{z(z - \cos \omega_0)}{(z^2 - 2z \cos \omega_0 + 1)}$$

$$\therefore Z[\cos \omega_0 n] u(n) = \frac{z(z - \cos \omega_0)}{z^2 - 2z \cos \omega_0 + 1}$$

✓ for $n \geq 0$
0 for $n < 0$.

ROC : $|z| > 1$.

5a

$$X(z) = \frac{1}{(1 - 2z^{-1})(1 - z^{-1})^2}$$

$$= \frac{1}{(1 - 2/z)(1 - 1/z)^2}$$

$$X(z) = \frac{z^3}{(z-2)(z-1)^2}$$

$$\frac{X(z)}{z} = \frac{z^2}{(z-2)(z-1)^2}$$

using partial fraction:

$$\frac{X(z)}{z} = \frac{A}{z-2} + \frac{B}{z-1} + \frac{C}{(z-1)^2}$$

$$A = \left| \frac{z^2}{(z-1)^2} \right|_{z=2}$$

$$= 4$$

$$A(z-2)^2 + B(z-1)(z-2) + C(z-2) = z^2$$

put $z = 1$

$$C(-1) \times 1$$

$$C = -1$$

put $z = 0$

$$4 \times (-1)^2 + B(-1)(-2) + (-1)(-2) = 0$$

$$6 + 2B = 0$$

$$B = -3$$

$$\therefore \frac{X(z)}{z} = \frac{4}{(z-2)} - \frac{3}{(z-1)} - \frac{1}{(z-1)^2}$$

$$X(z) = 4 \frac{z}{z-2} - 3 \frac{z}{z-1} - \frac{1}{(z-1)^2}$$

$\frac{z}{(z-1)^2}$
 $\frac{z}{(z-1)^2}$
 $\frac{z}{(z-1)^2}$

$$x(n) = \mathcal{Z}^{-1} \left[4 \frac{z}{z-2} - 3 \frac{z}{z-1} - \frac{z}{(z-1)^2} \right]$$

$$= 4 (2)^n u(n) - 3 (1)^n u(n) - n (1)^{n-1} u(n)$$

#

7

$\therefore x(n)$ has z-Transform.

5b

$$x_1(n) = \{1, 2, 3, 4\}$$

$$x_2(n) = \{0, 2, 4, 6\}$$

$$\text{Correlation of } x_1 x_2 = r_{x_1 x_2} = X_1(z) X_2(z^{-1})$$

$$\begin{aligned} X_1(z) &= \sum_{n=0}^3 x_1(n) z^{-n} \\ &= 1 + 2z^{-1} + 3z^{-2} + 4z^{-3} \end{aligned}$$

$$\begin{aligned} X_2(z^{-1}) &= \sum_{n=0}^3 x_2(n) z^n \\ &= 0 + 2z + 4z^2 + 6z^3 \end{aligned}$$

$$\text{now, } r_{x_1 x_2} = (1 + 2z^{-1} + 3z^{-2} + 4z^{-3}) (2z + 4z^2 + 6z^3)$$

$$= 2z + 4z^2 + 6z^3 + 4 + 8z + 12z^2 + 6z^{-1} + 12 + 18z + 8z^{-2} + 16z^{-1} + 24$$

$$= 6z^3 + 16z^2 + 20z + 36 + 22z^{-1} + 8z^{-2}$$

$$\therefore r_{x_1 x_2} = \{6, 16, 20, 36, 22, 8\}$$

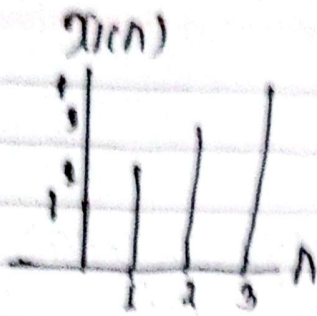


fig: $x_1(n)$

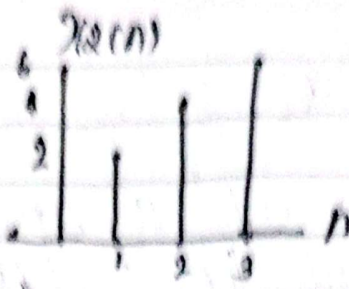


fig: $x_2(n)$

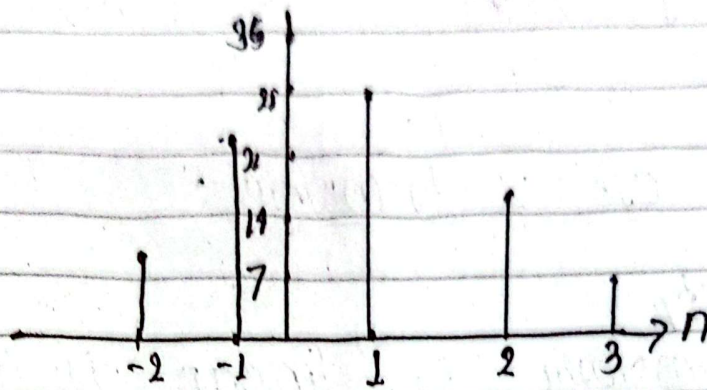


fig: $x_1 x_2$

7

6a

Z-transform of a discrete signal $x(n)$ is denoted by $Z[x(n)]$ and defined by as:

$$Z[x(n)] = X(z) = \sum_{n=-\infty}^{\infty} x(n)z^{-n}$$

Properties of z-transform:

① Linearity

A z-transform is linear if it satisfies:

$$x(n) \xrightarrow{Z} X(z)$$

$$a_1 x_1(n) + a_2 x_2(n) \xrightarrow{Z} a_1 X_1(z) + a_2 X_2(z)$$

where, a_1, a_2 is constant value.

if the condition not satisfies the z-transform not linear.

② Time Reversal

if $x(n) \xrightarrow{Z} X(z)$ in $r_1 < z < r_2$ then,

$$x(-n) \xrightarrow{Z} X(z^{-1}) \text{ in } \frac{1}{r_2} < z < \frac{1}{r_1}$$



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(iii)

Time shifting

$$\text{if } x(n) \xleftrightarrow{z} X(z)$$

let n_0 is some time shifting

then,

$$x(n-n_0) \xleftrightarrow{z} z^{n_0} X(z)$$

(iv)

Derivative

z^{-1}

$$\text{if } x(n) \xleftrightarrow{z} X(z)$$

$$\text{then } nx(n) \xleftrightarrow{z} -z \frac{dX(z)}{dz}$$

(v)

Constant Multiply

$$x(n) \xleftrightarrow{z} X(z) \quad r_1 < z < r_2$$

$$a^n x(n) \xleftrightarrow{z} a^{-1} X(z) \quad |a|r_1 < z < |a|r_2$$

(vi) Z-transform value of ∞ is not allowed

(vii) $Z(1) = \frac{z}{z-1}$

6b

$$x(n) = n^2 u(n)$$

$$x(0) = 0 \quad u(0) = 1$$

$$x(1) = 1 \quad u(1) = 1$$

$$x(2) = 4$$

$$x(3) = 9$$

$$\therefore Z[x(n)] = X(z) = Z[n^2 u(n)]$$

$$\sum_{n=0}^{\infty} x(n) z^{-n} = \sum_{n=0}^{\infty} n^2 u(n) z^{-n}$$

$$= 0 + 1z^{-1} + 4z^{-2} + 9z^{-3} + \dots$$

$$= z + \frac{4}{z^2} + \frac{9}{z^3} + \dots$$

6b

$$x(n) = n^2 u(n)$$

$$Z[u(n)] = \frac{z}{z-1}$$

$$Z[n u(n)] = -z \frac{d}{dz} \left(\frac{z}{z-1} \right)$$

$$= -z \left[\frac{(z-1) \cdot 1 - z \cdot 1}{(z-1)^2} \right]$$

$$= \frac{z}{(z-1)^2}$$

$$\therefore Z[n^2 u(n)] = -z \frac{d}{dz} \left(\frac{z}{(z-1)^2} \right)$$

$$= -z \times \frac{(z-1)^2 \cdot 1 - z \cdot 2(z-1)}{(z-1)^4}$$

$$= -z \frac{(z-1) [z-1-2z]}{(z-1)^4}$$

$$= \frac{z(z+1)}{(z-1)^3}$$

$$\therefore Z[n^2 u(n)] = \frac{z(z+1)}{(z-1)^3} \#$$

$$ROC = |z| > 1$$



7a) Region of Convergence

Region of Convergence is as short as
ROC is the set of values of z in z -plane that always provides finite value.

for example:

$$X(z) = \frac{z}{z-1}$$

$$\text{ROC} = |z| > 1$$

here $z \neq 1$ as it makes ∞ .

Properties of Z-transform: ROC

- ROC forms a ring of circle which origin is as center.
- ROC doesn't contain any place.
- Right sequence ROC doesn't contain infinity.
- Left sequence ROC not contained $z = 0$.
- ROC includes a circle. (4)

7b Energy and Power Signal.

Energy and Power signals are the two basic signals.

• Energy Signal

→ It has finite energy and power 0.

→ aperiodic in nature

$$\Rightarrow E = \int_{-\infty}^{\infty} x^2(t) dt$$

→ It is used to calculate energy.

$$\Rightarrow 0 < E < \infty$$

• Power Signal

→ practical periodic signal in nature.

→ finite power and infinite energy.

$$P = \frac{1}{T} \lim_{T \rightarrow \infty} \int_{-T/2}^{T/2} x^2(t) dt$$

→ It is used to find power.

$$\Rightarrow 0 < P < \infty$$